Express Riddler

1 April 2022

Riddle:

Multiplying polynomials can be challenging work. But every once in a while, there's an expansion that is particularly elegant. One of my favorites is the following expression:

$$(x-a)(x-b)(x-c)\dots(x-z)$$

The ellipsis indicates that every letter is subtracted at some point. If you multiply all these binomials together, can you write the expansion in simplest form?

Solution:

The 24th term in the expression is $(x-x)$, which of course is 0. Because all the terms are being multiplied, the solution is simply $\boxed{\mathbf{0}}$.